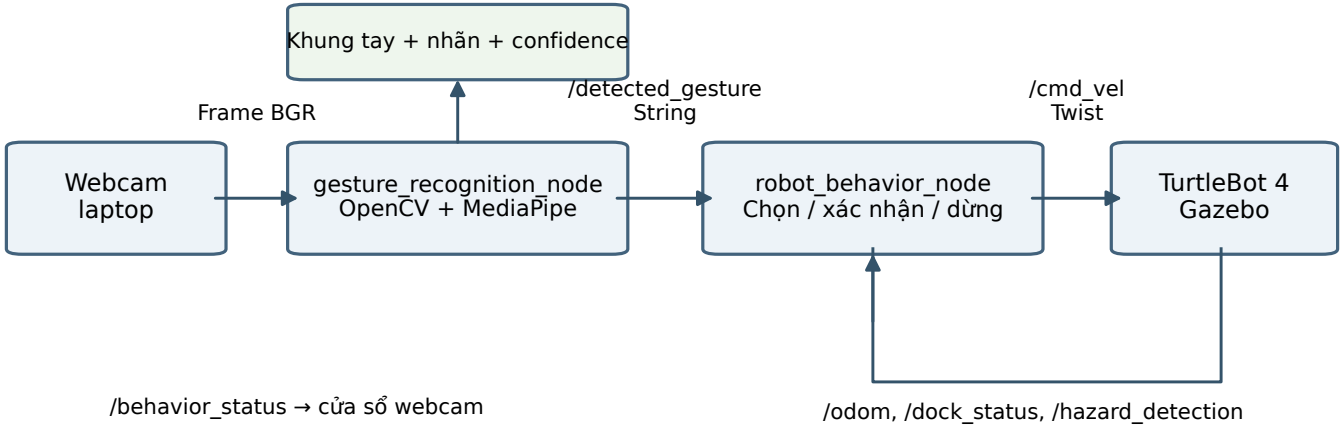
Khung tay + nhãn + confidence
Frame BGR
/detected_gesture
String
/cmd_vel
Twist
Webcam
laptop
gesture_recognition_node
OpenCV + MediaPipe
robot_behavior_node
Chọn / xác nhận / dừng
TurtleBot 4
Gazebo
/behavior_status → cửa sổ webcam
/odom, /dock_status, /hazard_detection